

Level of Traffic Stress and Infrastructure Network Analysis for Safe and Accessible Cycling

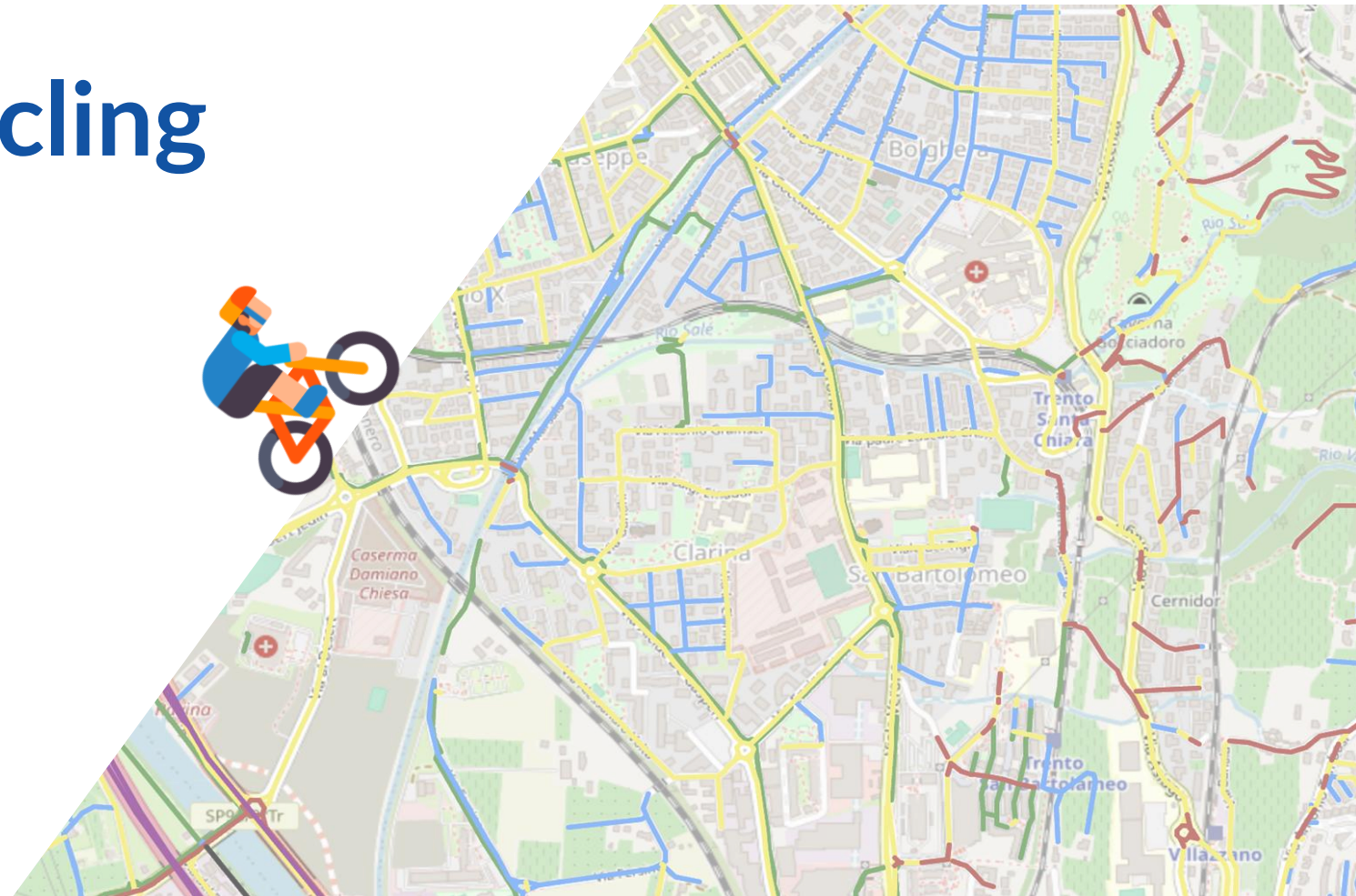


Leonardo Venturoso

Premio per la migliore tesi - Città di Bolzano, EUniverCities Network Conference, April 14th 2025



EUniverCities Network Conference
14 – 16.04. 2025, Bolzano Bozen



Background

We need to talk about
sustainability and bikes in
Alpine cities.



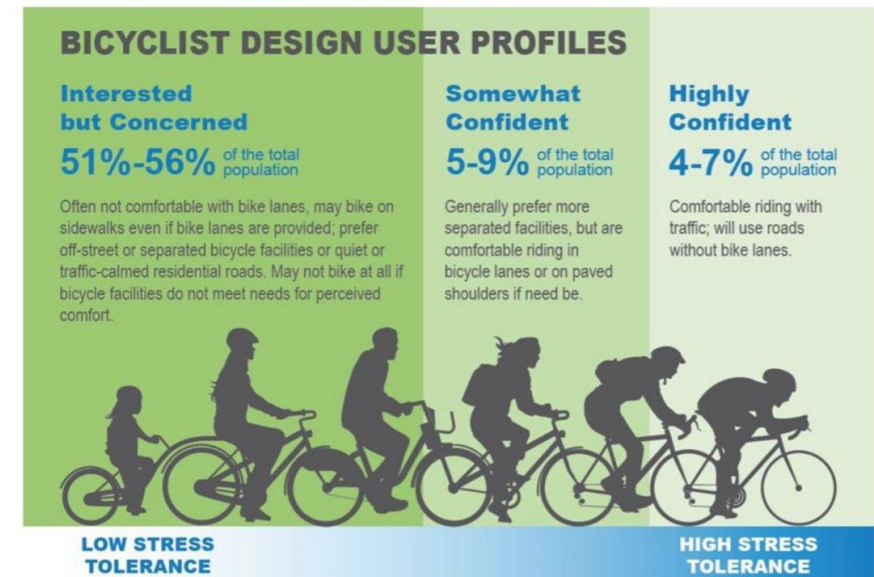
Background

How well the **bike network** in a community connects people with the places they want to go?



Background

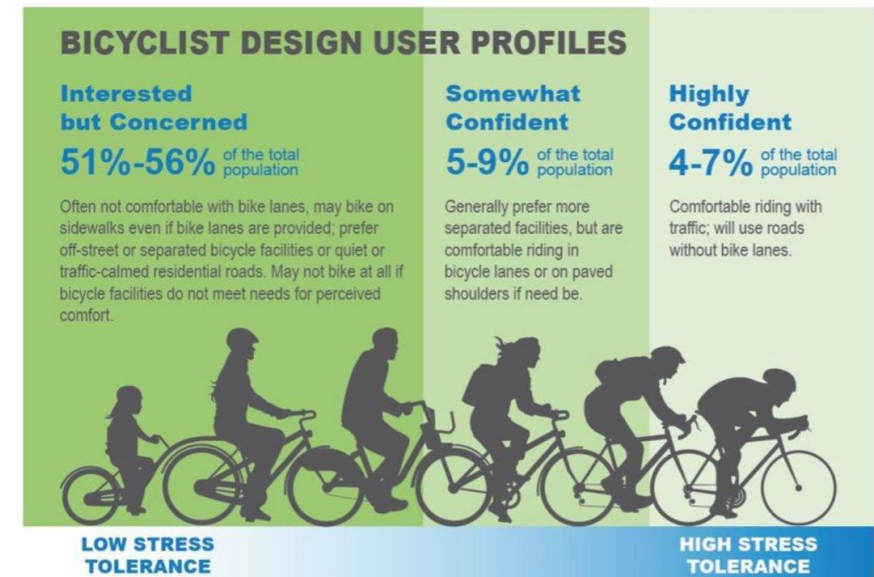
The **Traffic Stress** concept is a detailed metric used to evaluate *the range of difficulties and discomforts encountered by cyclists on specific road segments. It encompasses both the immediate feelings of discomfort, anxiety, or danger experienced by cyclists, as well as the wider effects of factors such as infrastructure design, vehicle behavior, environmental conditions, and topography**.



* Winters, M., Davidson, G., Kao, D., & Teschke, K. (2011). Motivators and deterrents of bicycling: comparing influences on decisions to ride. *Transportation*, 38, 153-168.

Background

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Research Goals

1. Investigate the **factors influencing the choice between bicycles and cars**, integrating aspects such as logistical and environmental conditions.
2. Focus on **quantifying the relationship between traffic stress levels and risk perception**, considering infrastructural and environmental conditions (like centrality measures and network gaps) and qualitative factors (like accessibility and accident presence).
3. Develop a **Practical Open Digital Tool (LTS-Bike Plan)** to aid public administrations by providing insights for potential interventions to increase bicycle usage.
4. Utilize **Trento and Bolzano as case studies** due to familiarity with these areas for effective research validation and their **geomorphological variability (flat vs challenging terrains)**.

Research Questions

- 1) In urban landscapes with diverse geomorphological characteristics, **which streets or areas exhibit high LTS**, and which ones are characterized by low stress?
- 2) **Which streets and areas in the urban landscape exhibit higher BNA scores** and are near common destinations, including services and amenities?
- 3) Are the networks composed of subgraphs distinguished by different levels of traffic stress? If this is the case, **how many high-stress gaps occur within different areas of low stress?** Moreover, which of these high-stress gaps are most **crucial**?

One of the **future goals** of the work of Vybornova, A., Cunha, T., Gühnemann, A., & Szell, M. (2023). *Automated detection of missing links in bicycle networks*. *Geographical Analysis*, 55(2), 239-267.



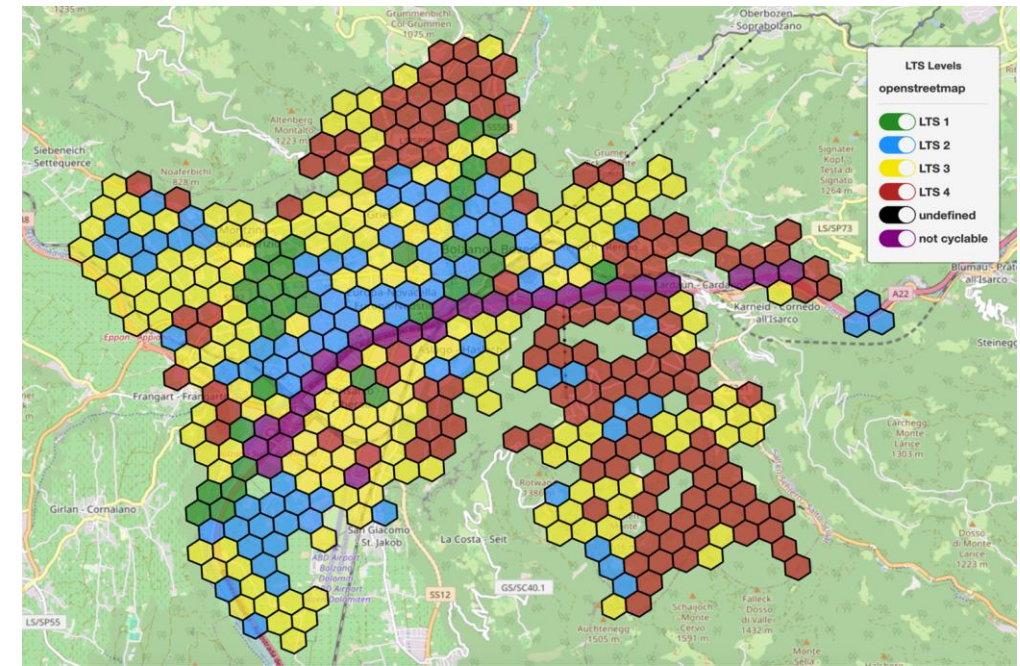
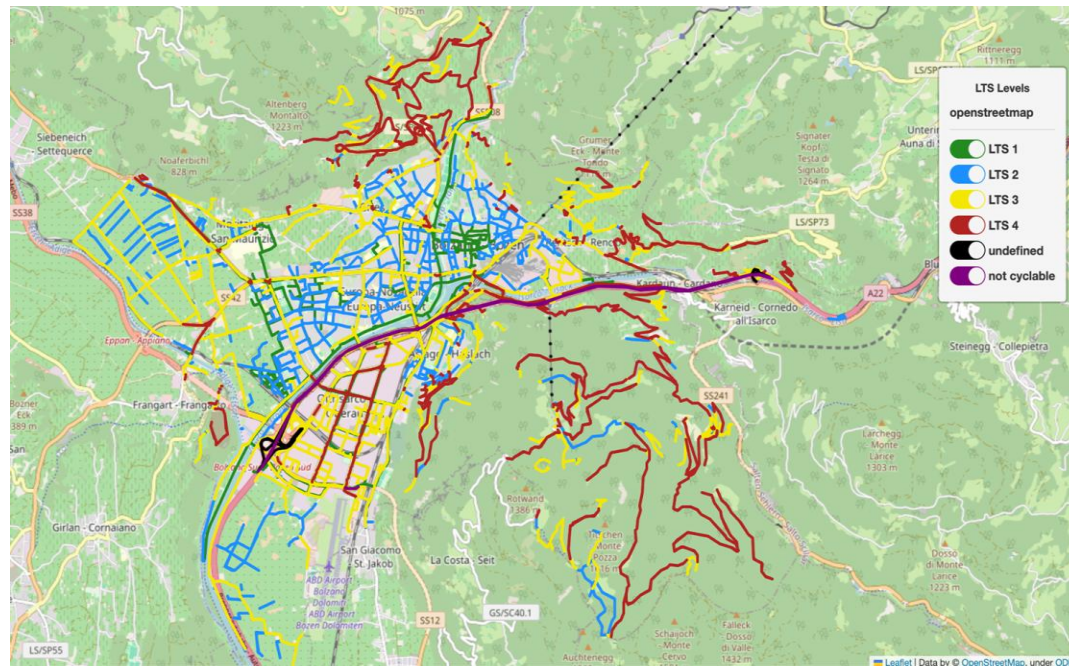
Research Questions

- 4) Does a **spatial relationship** exist between the **locations of accidents** and the **high traffic stress gaps** within the urban network?
- 5) Is it possible to **predict areas at high risk for accidents** by combining quantitative metrics, such as centrality measures, with qualitative assessments like accessibility scores and levels of traffic stress?



Results

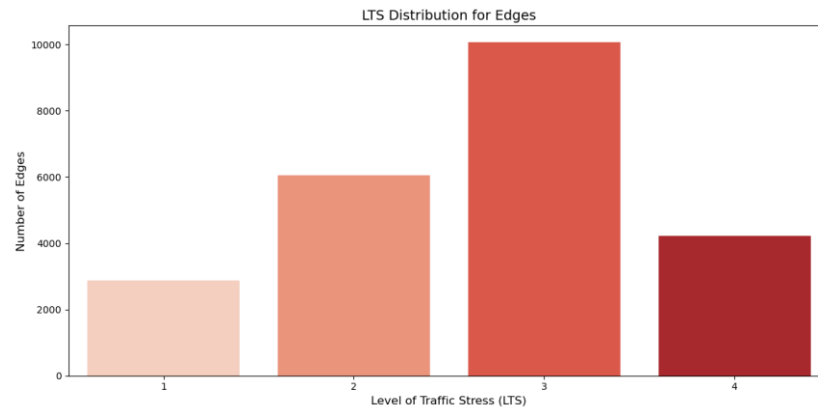
Adaptation of the LTS Scoring Process (Bolzano) - LTS map



Results

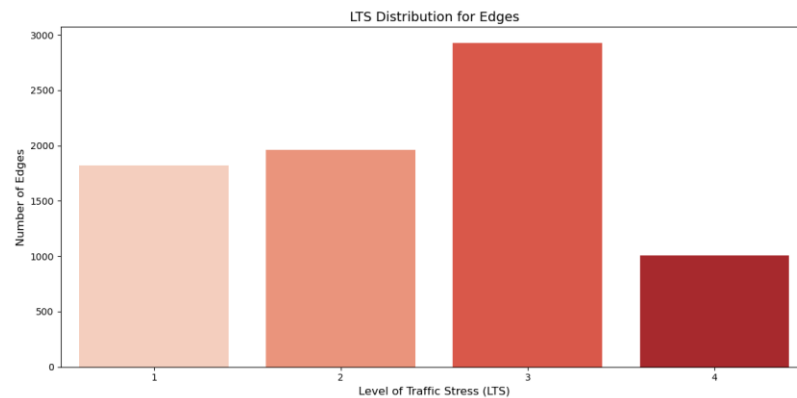
Infrastructure and Network Analysis - LTS Distribution

Trento



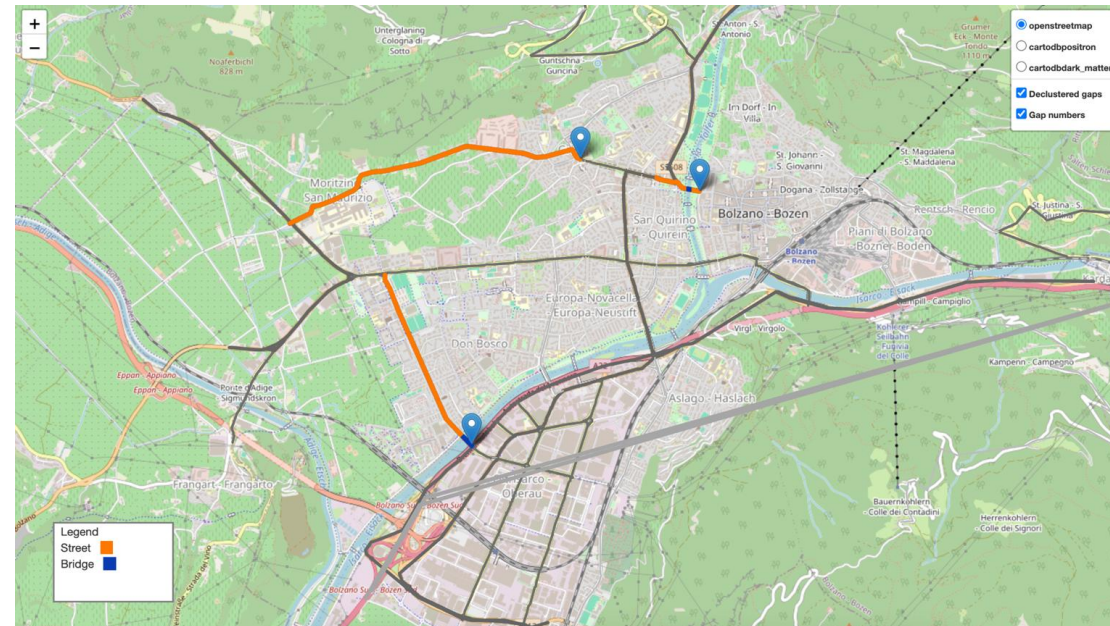
The examination of LTS level distributions showed comparable patterns across both case studies, with **LTS 3** emerging as the **predominant level**. Specifically in the case of Bolzano, it was observed that LTS 1 and 2 exhibit a notably similar presence.

Bolzano



Results

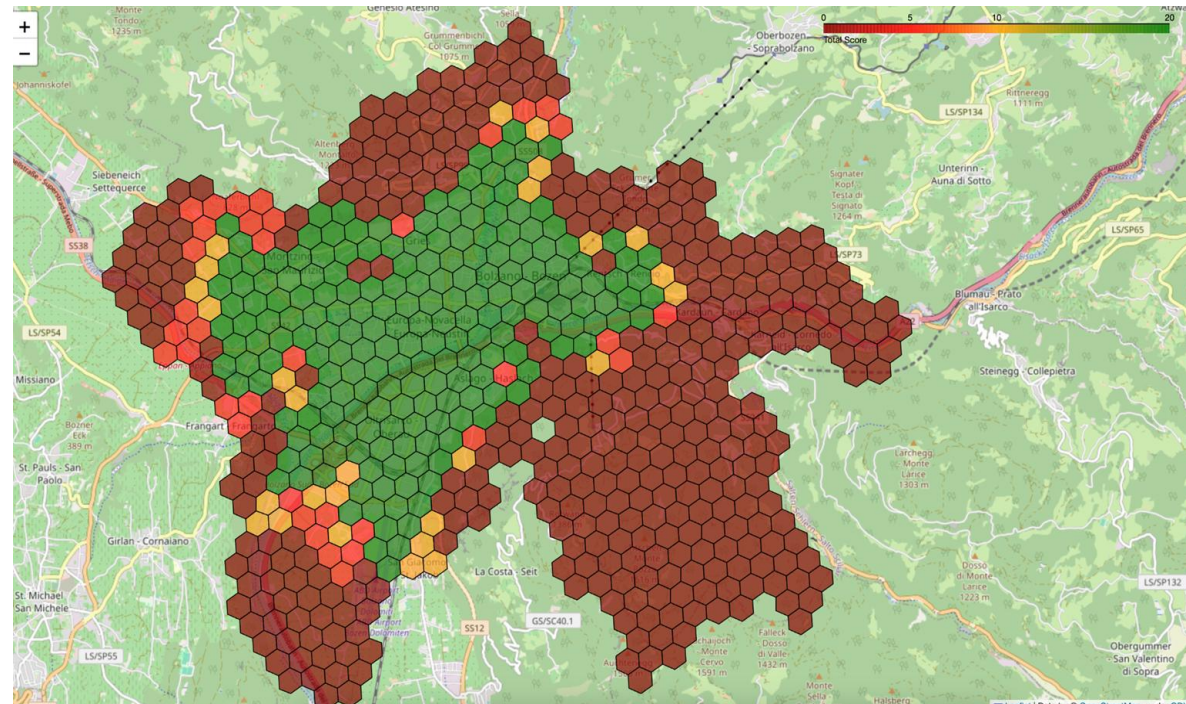
Infrastructure and Network Analysis - Gap Analysis (LTS IPDC procedure)



Bolzano

Results

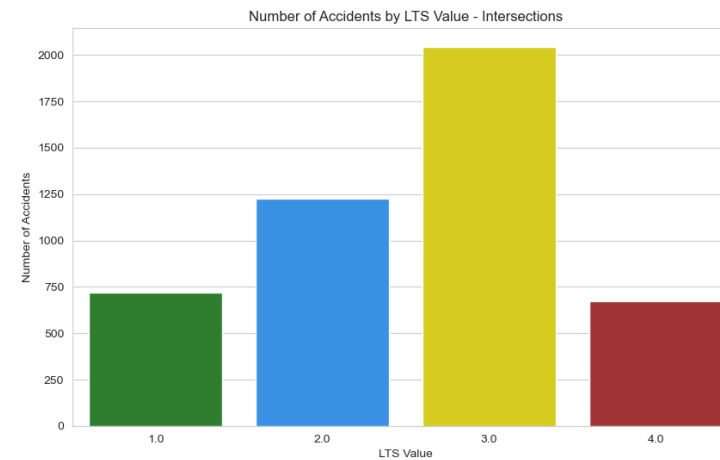
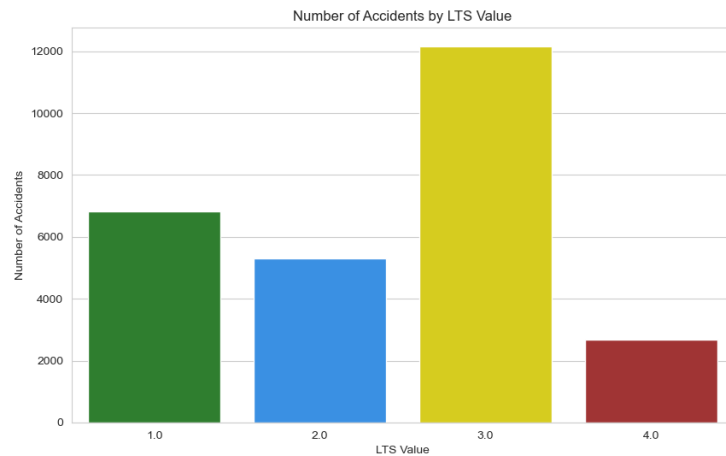
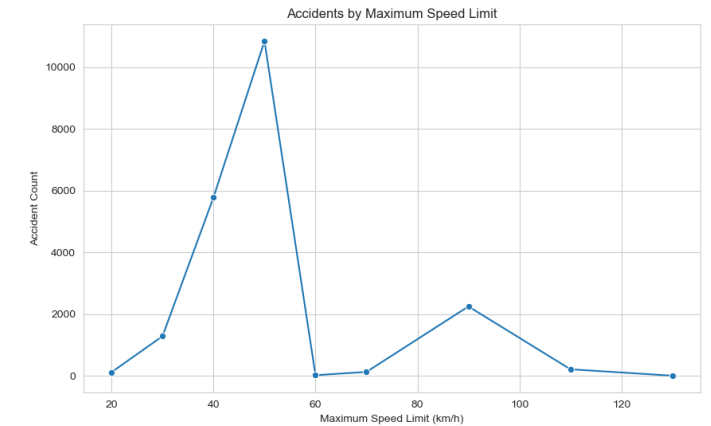
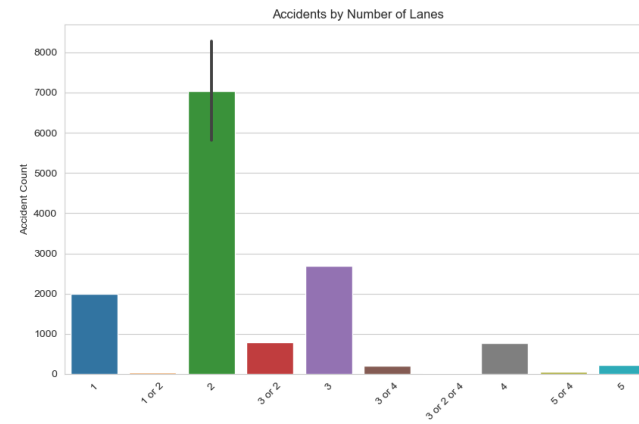
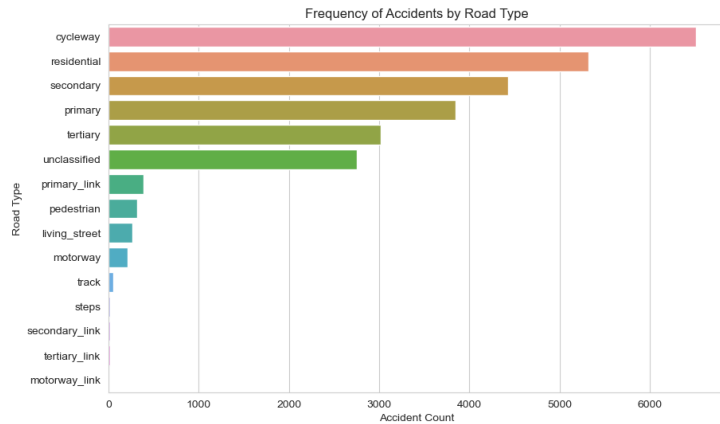
Infrastructure and Network Analysis - BNA Score (People for Bikes)



Bolzano

Results

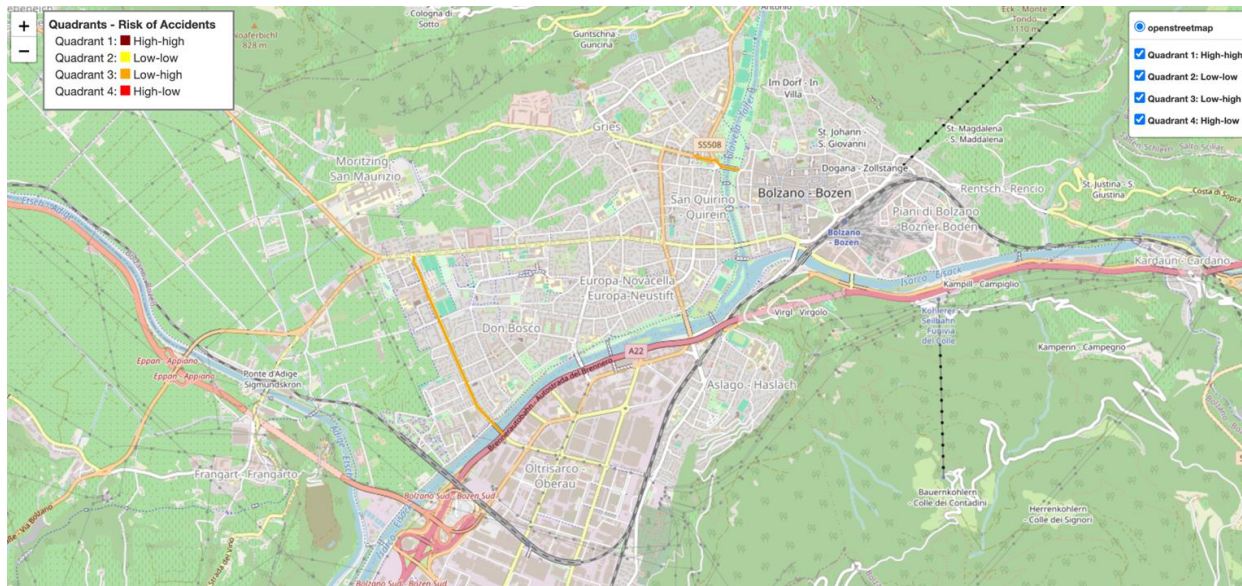
Infrastructure and Network Analysis (Bolzano) - Accident Analysis



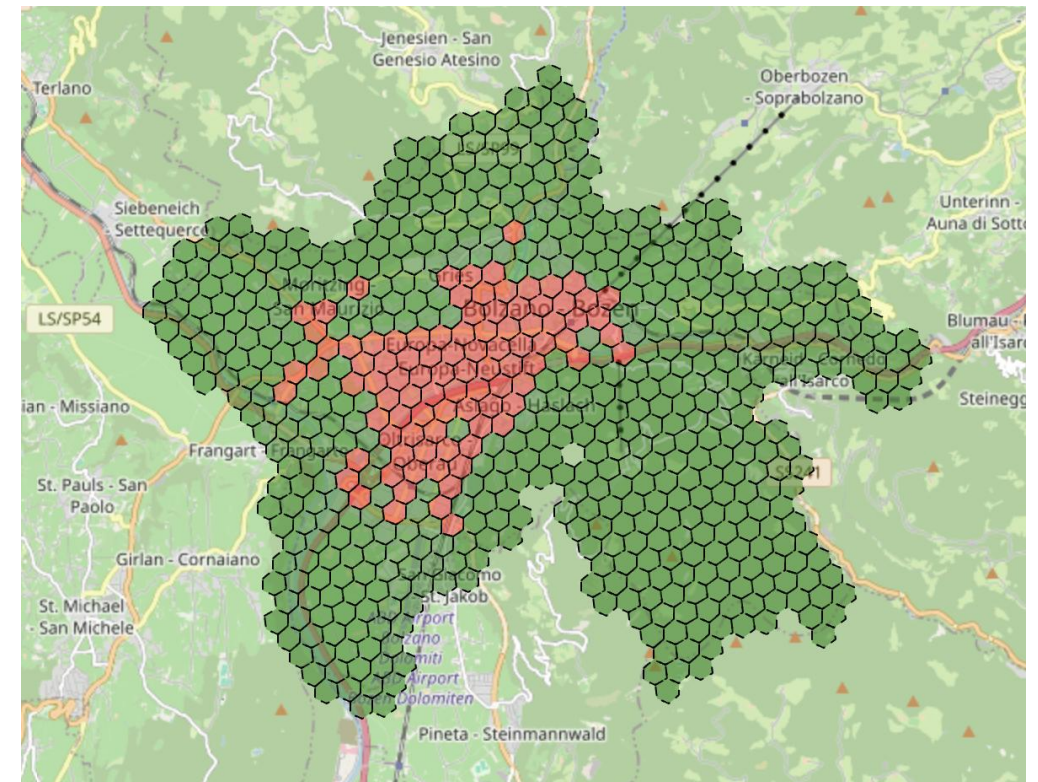
Results

Final Steps Analysis (Bolzano)

Spatial Relationship Analysis



High Risk Areas Prediction



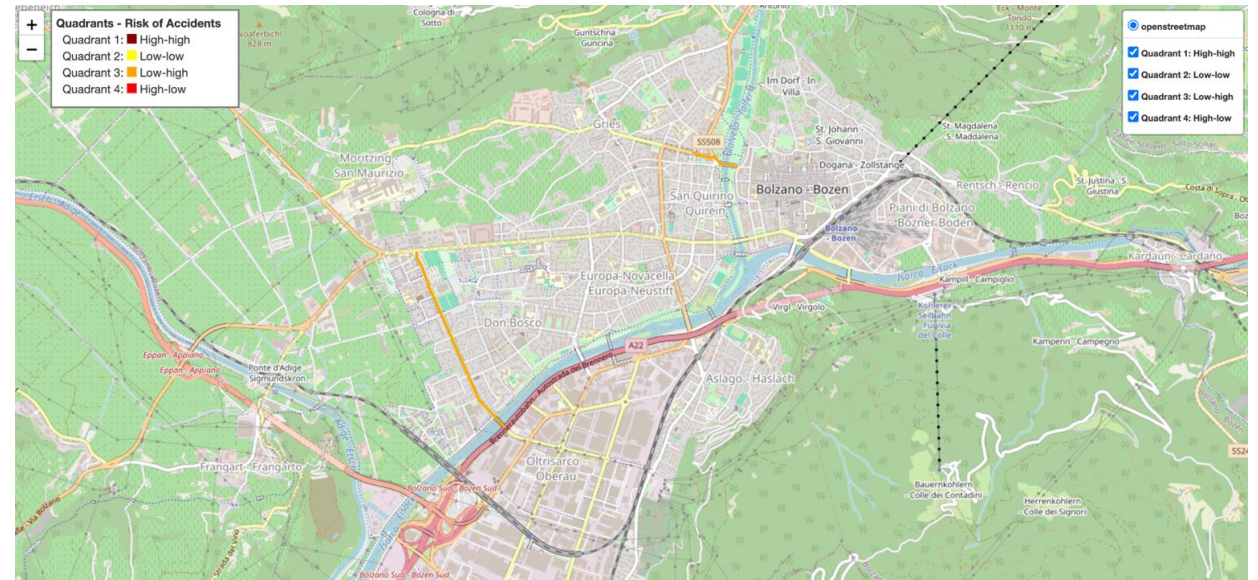
Results

Final Steps Analysis (Bolzano)

Risultati

Quadrant	Description	Number of Components
1	High-High (HH)	0
2	Low-Low (LH)	0
3	Low-High (LL)	2
4	High-Low (HL)	0

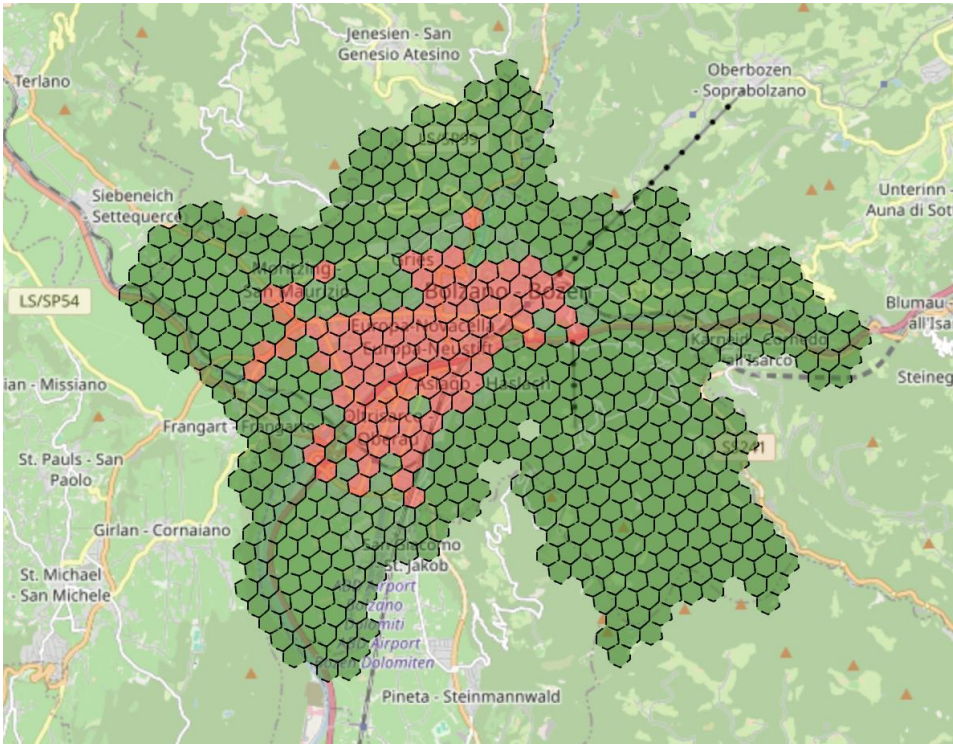
Spatial Relationship Analysis



Results

Final Steps Analysis (Bolzano)

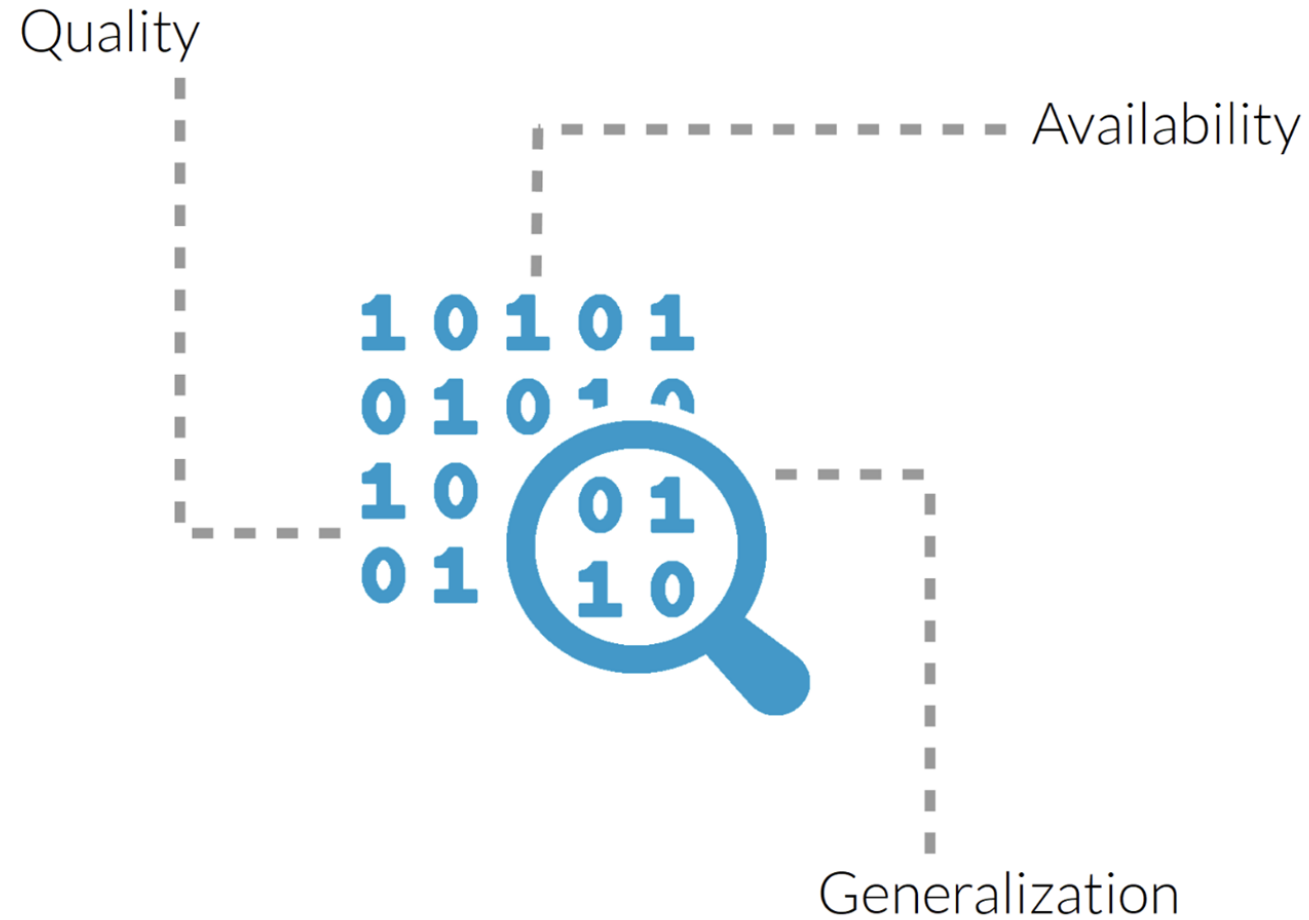
High Risk Areas Prediction



Risultati

Metric	Random Forest Classifier
Accuracy	0.92
Macro Avg F1-Score	0.91
Precision-Recall AUC	0.93
ROC-AUC Score	0.96

Limitations and Future Research



Conclusions

- 1) In urban landscapes with diverse geomorphological characteristics, **which streets or areas exhibit high LTS**, and which ones are characterized by low stress?
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Thank you for your attention